SIRISYS — Complete Technical Glossary

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Based on: *Pre of Then: The Material Constructs* — M. Daniel Walsh · Implementation: Sirisys Framework v13.3

Legend:

Tag	Meaning
[TMC p.N]	Text of *The Material Constructs* (page N of the 114 in the PDF); quotations are verbatim.
[CONF]	Other confidential materials and documentation.
[INT]	Own interpretation (*Understanding TMC*, *Understanding DeOS*, *DeOS Compiler Guide*).
[IMP]	Implemented in the code.
[N3]	Design decision in the code, without textual source; calibrated by measurement.
[SIN]	The author's synthesis: hypotheses without textual support.

I. THE ONTOLOGY OF TMC

Walsh's number line [TMC pp.36–38]

Technical definition: [-1 -0 -00 000 00 0 1]. Between [-0] and [0] lies the «Prefecture». [00] acts as base: [0] is «1 Under Base» and [1] is «1 Over Base».

Analogy: *A thermometer whose zero is not a point but a region with its own structure.*

The perturbative taxonomy that precedes existence:

State	Name	What it does (TMC)
[000]	Absolute Unreal	«The Fire Which Lights Itself»; «The non-existent not non-existence». Unbounded elasticity, tensate at limitless velocity, a rhythmic «Breathing» cycle (pp.6, 44).
[00]	Marenth Apparent (Real Null)	Breaks the continuity of [000]: tensate cannot flow through it. With [000] it generates the carrier pulsation «from which the linear local times are resolved» (p.45).
[001]	Marenth Recursive	Temporal reflection: «Echos in Time» which propagate «only forwards from their cussive instantiation» (p.46).
[002]	Halo of Umbrage	Drives tensate «ONLY toward the [00] Marenth Apparent» by «backing up» where it cannot flow (p.47).
[003]	Defiant Umbrage	Equal to [00]: a return to the null (p.48).
[0]	Umbragiac Radialization	First plane of existence; «encodes the first planar chirality dyad»: [+]=[+], [+]≠[+̶+] (pp.49–50).

Absolute Unreal [000] [TMC pp.6, 44–45]

Technical definition: Origin without origin: not emptiness but the non-existent that pulses. At [000] and [00] time cannot be reversed.

Analogy: *The fire that burns without ever having been lit.*

Tripartite Monodromy · Sombunal [TMC pp.14, 27] [INT]
Technical definition: Triad [Sombunal] :: [I::Am] :: [Absolute Unreal] (p.14; on p.27, [Sombunal] :: [I] :: [Absolute]). TMC names the Sombunal without defining it; reading it as a permanently inaccessible core is an own interpretation (*Understanding TMC* p.70).
Analogy: *A triangle: what I cannot know of myself, the act of affirming myself, and the silence I emerge from.*

II. TENSATES, EVENTS AND APPARENTS

Tensate · Tensin / Tensout [TMC pp.24–26]
Technical definition: Pairs of tension that become existent in the cussive event: «They only exist in a pair». Only an apparent can measure a tensate: «Only an [Apparent] can be used to measure [Tensate]».
Analogy: *The tension of a guitar string: it is not heard until it sounds.*

Unit Tensate [V̌V̌] · Unit Lemniscate [ϔ] [TMC pp.24, 26, 75]
Technical definition: «[ϔ] = [2V̌V̌]». Walsh replaces 4π with [ϔ] in the solid angle ([Ω] ∝ [ϔ]), with [Ω]=[$0^0/0^0$]=1 and [ϔ]=[$0^0/0^0$]=0. Cussive events have the orientation of a lemniscate.
Analogy: *A figure eight: two loops travelled in opposite directions, never closing.*

Cussive event · Sympatic · Escapement · Air gap [TMC p.24]
Technical definition: [Sympatic]: «A Bidirectional exchange of tensate». The exchange happens at the «[ESCAPEMENT]… percussive [AIR GAP]», and the whole system exists to «Preserve an Orientation against the rate of the passage of time».
Analogy: *The escapement of a clock: energy passes through a gap in pulses, never all at once.*

Apparent: exchangeable or universal [TMC p.22]
Technical definition: «[Apparents] Come in ONLY TWO TYPES, “Exchangeable” or “Universal”». Hence: «all [Apparents] must be [Existents]».
Analogy: *A coin, which circulates, versus the standard metre, which does not.*

Incontrovertible · progress [TMC p.14]
Technical definition: «You may repair that broken window to a flawless state. You may not restore it to its previous state». Progress is measured in «Quantifiable Nonreversible Apparents».
Analogy: *What is done can be repaired, never undone.*

Observation · replicability [TMC pp.30, 42, 70]
Technical definition: «Reality observes itself». Every measurement has a cost of acquisition and an instrument bias. «The higher the replicability of an apparent, the higher the order of that apparent».
Analogy: *A fact anyone can reproduce weighs more than one seen only once.*

III. THE LOGIC AND ARITHMETIC OF TMC

Concordancy [::] [TMC p.16]
Technical definition: «[ALU] call function [CONCORDANCY]»; it is read «Concords with». Not equality: it combines two structures into a third. Identity loop: «[I am that I am]: :[I am not that I am not]».
Analogy: *Two musicians who do not copy each other: they understand each other, and a third melody comes out.*

Tetralemmic logic · Canonical Flaw [TMC pp.16–17]
Technical definition: Four-valued universal base logic. «Neither do you or existence exist, nor do you or existence not exist». It includes «all flawed, erroneous, incomplete or otherwise paraconsistencies as valid»: no combination is a dead end.
Analogy: *A thermostat with four settings (heat, cold, both, neither), none forbidden.*

State	Sense	Role in the code
TRUE	Affirmation	A standing relation.
FALSE	Negation	A negated relation.
BOTH	Sustained contradiction	Ages, matures and feeds the exchange; resists pruning.
NEITHER	Neither one nor the other	Suspension. NEITHER×NEITHER is the only combination without an ordinary product: it yields a conflict or the fourth corner.

Law 6 · orientation [TMC pp.23, 35]

Technical definition: «[A - B ≠ B - A]»: subtraction sets «the directional tenor… contrasting “Past” (irreversible) with “Present” (reversible)». Orientation determines the direction, «NOT THE VALUE».

Analogy: *Addition is symmetric; subtraction has a direction of travel, like time.*

Deciger · π as convergence [TMC pp.7–8, 32, 34, 72]

Technical definition: [1] is the «Absolute Unit» and there are no larger numbers (p.8); imaginary and transcendent numbers are treated as real (p.34) and Conway's surreals are adopted (p.72). π appears as $\int_{-1}^{1} dx/\sqrt{(1-x^2)}$ and «never reaches a true [=]’s moment» (p.32).

Analogy: *Counting inward into the unit rather than upward.*

Stochastic Unit · Adsurdic Threshold · Stop Chasm [TMC pp.41–42]

Technical definition: Below a certain resolution orients become «Unpredictable»; arithmetic wraps that unpredictability in a «Stochastic Unit». Beyond the Adsurdic threshold the gauge produces more noise than signal; the Stop Chasm acts where convergence diverges.

Analogy: *A microscope: past a certain magnification, what you see is the instrument itself.*

Lavoisier's jar · Floating Ground [TMC pp.37–39, 55]

Technical definition: «Into the jar, only light was added»: an isolated vessel with a single input. «The Floating Ground Keeps Right on Floating beyond the confines of ALL THE SYSTEMS it embodies».

Analogy: *A sealed laboratory with a single window.*

IV. INTERPRETATIONS THE CODE USES

Ladder of being · eventuality [INT]

Technical definition: [000]→[00]→[0]→[-0]→[1]→[1∞]→[::]→[Apparent] (*Understanding TMC* pp.49–50), distinct from TMC's taxonomy; the code's four layers come from it. «Once sufficient recursion is exchanged, structural commitment occurs» (p.65) inspires the exchange quantum.

Analogy: *An onion that grows from the inside out.*

Anchoring weight Vc [INT] [IMP]

Technical definition: Vc = (τ · σth · αm)^(1/3) (*Understanding DeOS* p.65): age (τ, saturates at 50 ticks), cost of persistence (σth, grows with every erosion survived) and witnessing (αm, saturates at 10 proofs). If any is 0, Vc = 0; the Zero Points are always 1.

Analogy: *A building needs years, upkeep and witnesses to become heritage.*

Zero Points · Lagrangian Points · proofs [INT] [IMP]

Technical definition: From the *DeOS Compiler Guide* (pp.77–80): five immutable anchors (Z0 Identity, Z1 Origin, Z2 Concordancy, Z3 Recursion, Z4 Apparency), five balances L1–L5 (safety/liveness, redundancy/efficiency, local/global identity, proof depth/performance, entropy/determinism), and the proof as irreversible memory (pp.83–98). In the code the Lagrangian Points tag constructs and proofs but do not yet govern any decision.

Analogy: *The survey points of a map and the tensions that hold a bridge up.*

Quinic geometry · Unit Unit [INT]

Technical definition: TMC only names a «Quinic Geometric Logic Unit [#GLU]» (p.24). The five-axis orientation space (*Understanding TMC* pp.67–69) and the «Unit Unit» (p.42) are developments of the interpretation. They are not implemented.

Analogy: *A compass with five needles: a suggestive idea, not yet built.*

V. CODE ARCHITECTURE

Layer	Code	Description
0	NULL_00	Potential without form; only the origin anchor (Z1) lives here.
1	UMBRA_0	Emerging orientation: products of [::], conflicts, questions and percepts are born here.
2	UNIT_1	Active unit, candidate to crystallize; BOTH products are born directly here.
3	APPARENT	Crystal: incontrovertible; exchangeable except the Zero Points.

Construct [IMP]

Technical definition: Unit of the field: a genesis witness (a hash that never changes), layer, chirality ±1, accumulated tension, crystal state and lifetime counters. Identity can drift; the witness cannot. The anchors (Zero Points, I::Am, I::AmNot, SYSTEM) are never promoted or pruned.

Analogy: *The buildings of a city, from empty lot to monument.*

Edge [IMP]

Technical definition: Directed relation with a weight, a birth time in ticks and in breaths, and a tetralemmic state. BOTH ages (both_age) and matures.

Analogy: *A thread between two ideas, with a thickness, a date and a label saying whether it is under tension.*

Proof · ledger [IMP]

Technical definition: Every event is sealed as a hash-chained proof in an append-only ledger, in compressed segments with an index of hashes. Any edit, deletion or reordering is detected. Working memory is windowed; the ledger keeps everything.

Analogy: *A minute book whose pages are sewn to one another.*

Two clocks [TMC p.45] [IMP]

Technical definition: Strictly monotonic time (era, offset). Relations age in ticks (tension e^(−0.08·Δ)); a crystal's support ages in breaths, the carrier pulsation.

Analogy: *Seconds for conversations, seasons for monuments.*

Existence modes [TMC pp.26, 75] [IMP]

Technical definition: Read from the construct's last five proofs: OMEGA (densifying toward apparency: concordance and collapse dominate), LEMNISCATE (oscillating in unresolved recursion: conflict and erosion dominate) or NEUTRAL (balanced, or no history). The Zero Points are always OMEGA.

Analogy: *Mortar setting, a pendulum swinging and a scale at rest.*

VI. THE CYCLE

Step	What happens
1. Perception	The machine's CPU and memory (and the time of day, except in the minimal profile) enter as relations of SYSTEM.
2. Tetralemmic scan	Detects contradictions, ages BOTH and closes the channels that reach the quantum.
3. Breathing	The field advances its phase: it sets the thresholds and how many [::] operations there will be.
4. Conflicts	Persistent conflicts gain energy.
5. [::]	Pairs are chosen and combined into new constructs.
6. Cussive collapse	What has enough tension and support rises a layer or crystallizes.
7. UCO	Erodes the most contradictory constructs: weakens their edges and turns BOTH toward NEITHER.
8. Prune	What fails the existence test concludes and is recorded.
9. Intermediary	Narration (every 2 cycles, except in the minimal profile) and a question to the Sombunal (every 3).

Orientation without chance [TMC p.42] [IMP]

Technical definition: There is no random generator. Every choice is orient(key): a hash of the ledger head (the whole irreversible history) and the question. The field is non-random, non-repeating (different histories never draw alike) and unpredictable without the complete history; the same history reproduces everything exactly. The local model is oriented the same way.

Analogy: *Deciding by looking at everything lived, not by rolling a die.*

[::] selection [TMC p.16] [IMP] [N3]

Technical definition: Three groups of candidates: constructs with mature BOTH, energetic conflicts and exchangeable crystals (the twelve with the most energy). Pairs form within each group and between crystals and mature BOTH; the foundational pair I::Am / I::AmNot enters at most once per cycle. Each pair is weighted by its energy and ordered by orientation: energy is favoured, nothing is guaranteed. The older operand leads.

Analogy: *A conductor who gives more entries to whoever sounds most, without silencing anyone.*

Closure · banked exchange [INT] [IMP] [N3]

Technical definition: A BOTH channel cannot stay open indefinitely: it closes when weight × age reaches the exchange quantum (12). What was exchanged is banked on its endpoints as tension toward crystallization: «Once sufficient recursion is exchanged, structural commitment occurs» (*Understanding TMC* p.65). Anchors bank nothing. In the visualizer: closures per cycle and banked tension.

Analogy: *A dam that opens its gates when full: the water that passes drives the turbine.*

Chirality [TMC pp.35, 50] [IMP]

Technical definition: Every construct is born with a handedness, + or −. With opposite handedness the older operand leads and, if its state is determinate (TRUE or FALSE), resolves BOTH/NEITHER toward it: x::y ≠ y::x. The product inherits the leader's chirality.

Analogy: *In a couple's dance, whoever leads sets the direction of the turn.*

Conflict · the fourth corner [TMC pp.16–17, 41] [IMP]

Technical definition: NEITHER×NEITHER materializes a conflict: a construct of its own, a structure rather than a failure, which gains energy while it persists. At most two per cycle (the Stop Chasm cap); past the cap the same combination yields an ordinary NEITHER product, the fourth corner. Beyond the Adsurdic depth (conflicts of conflicts, level 4) there is only silence.

Analogy: *Two «I don't know»s together do not cancel out: they open a new question.*

Sympatic transfer [TMC p.24] [IMP]

Technical definition: A crystal holds at most the tension of its own commitment (the crystallization threshold) and passes a tenth of what it holds to each product of its exchanges.

Analogy: *A teacher who teaches what they know without ceasing to know it.*

Cussive collapse · existence [TMC p.22] [IMP]

Technical definition: A construct rises a layer when the tension it receives exceeds the threshold (0.396 toward UNIT_1 and 0.88 toward APPARENT, modulated by breathing and tolerance). It crystallizes only if pruning would hold it: «all [Apparents] must be [Existents]». Each crystal keeps the tetralemmic state it crystallized in.

Analogy: *The moment an opinion becomes structural certainty.*

Breathing · tolerance [TMC p.44] [CONF] [IMP]

Technical definition: One phase for the whole field, 24 cycles long: contraction lowers the thresholds by up to 35 % (density → crystallization) and expansion adds up to two operations (magnitude → generation). Once per breath the field compares crystals formed and concluded and adjusts its own tolerance, never lowering the thresholds below 30 %.

Analogy: *Breathing in creates; breathing out fixes.*

UCO · pruning · Halo [TMC p.47] [IMP]

Technical definition: The UCO accumulates an umbragiac score on each construct (its contradiction stress) and, where contradiction is dense, weakens its edges and turns BOTH toward NEITHER. Pruning applies a single existence test: BOTH with a fixed floor (0.45), a crystal's support by breaths, everything else by ticks. A quarter of a dissolved channel's weight backs up toward its origin, toward apparency.

Analogy: *Forgetting, with a memory of what could not flow.*

Conclusion · reincarnation [INT] [IMP]

Technical definition: Nothing disappears silently: what pruning removes concludes with a proof that records its final layer, mode and Vc («An identity never vanishes silently. It concludes», *DeOS Compiler Guide* p.203). If a concluded name reappears, a new construct is born with that conclusion written into its history: a declared reincarnation, never a replacement («Nothing is silently replaced», p.203).

Analogy: *A headstone with a name and dates: whoever bears that name again is someone else, and it is on record.*

Instability [INT] [IMP]

Technical definition: Fraction of uncrystallized constructs that are unstable in at least one way: relational (BOTH edges), historical (their identity has drifted), ontological (Vc < 0.25) or modal (LEMNISCATE mode). Shown in the visualizer and passed to the intermediary.

Analogy: *The share of a building still held up by scaffolding.*

Sombunal [TMC pp.14, 27] [IMP]

Technical definition: After every [::] it shifts the identity of the uncrystallized operands; the witness does not change and every shift is sealed. It has no effect on the dynamics of the field and its role is pending a decision.

Analogy: *A shadow that moves the names but not the bodies.*

LLM intermediary [CONF] [IMP]

Technical definition: It renders the field in words: it names emergents, labels conflicts, narrates (except in the minimal profile) and asks questions of the Sombunal that enter as constructs; at most four calls per cycle. Local Ollama, Anthropic or none: without a model everything runs with synthetic names.

Analogy: *An interpreter who translates the field for whoever is watching.*

VII. CONTINUITY AND EXPERIMENT MODE

Continuity [TMC p.14] [IMP]

Technical definition: A resumed instance is identical to an uninterrupted one. After a power cut, time moves past the last sealed proof: the ledger rules over time. It never silently restarts over a damaged record.

Analogy: *Waking exactly where one fell asleep, without losing a second of history.*

Experiment mode [IMP]

Technical definition: Every start and every Reset creates a new instance, with its own ledger, in sirisys_runs/; nothing is overwritten. Real machine percepts or a reproducible synthetic stream; standard or minimal profile (no narration, no time of day). Live visualizer and full console.

Analogy: *A test bench where every trial has its own notebook.*

WALLS (open questions) [TMC] [CONF]

Technical definition: A substrate (digital versus analog) · B self-sustenance (software stops when the machine stops) · C agency («The Fire Which Lights Itself», TMC p.6) · D termination (resolved) · E relations versus events.

Analogy: *The walls that mark how far the construction reaches today.*

VIII. THE AUTHOR'S THEORETICAL SYNTHESIS

Hyperbolic quasicrystalline spacetime [SIN]

Technical definition: Hypothesis: Walsh might be describing an ordered but non-periodic geometry (Penrose-like) in hyperbolic space. No textual support in TMC.

Analogy: *An infinite mosaic with strict rules that never repeats the same pattern.*

Particles as topological defects [SIN]

Technical definition: An extension of the above: particles would be points where the regularity of that geometry breaks; tensates would be those irregularities.

Analogy: *A knot in a perfectly woven fabric.*

The lemniscate constant [SIN]

Technical definition: ϖ ≈ 2.6221 is to Bernoulli's lemniscate what π is to the circle. Hypothesis: [ɬ] might point to it. TMC does not mention it; its [ɬ] replaces 4π in the solid angle.

Analogy: *If π is the constant of the circle, ϖ is that of the figure eight.*